

1 Single-Scale Alignment

Exhaustive translation search on the three low-resolution JPEG plates

The blue plate is fixed while green and red are searched independently over every integer displacement in $[-15,15]^2$. Scores use only the common non-wrapping overlap, so pixels shifted off one edge never reappear on the other.

1.1 L2 and NCC

metric	objective	strength	role
L2 / normalized RMSE	minimize $\sqrt{\text{mean}((z(B)-z(M_d))^2)}$	Direct pixel error; lower is better	More sensitive to local cross-filter brightness
NCC	maximize $\text{mean}(z(B) z(M_d))$	Mean/scale invariant; higher is better	Primary similarity signal for JPEG search

Here $z(.)$ denotes standardization over the same overlap. Normalized L2 is retained as an interpretable audit, while NCC is more stable when historical filter exposures differ.

1.2 Measured JPEG baseline

JPEG	G -> B	R -> B	NCC G / R	L2 G / R
cathedral	(5, 2)	(12, 3)	0.95237 / 0.90071	0.30866 / 0.44563
monastery	(-3, 2)	(3, 2)	0.8405 / 0.66113	0.56481 / 0.82325
tobolsk	(3, 2)	(6, 3)	0.92034 / 0.85351	0.39915 / 0.54128



cathedral - exhaustive single-scale result

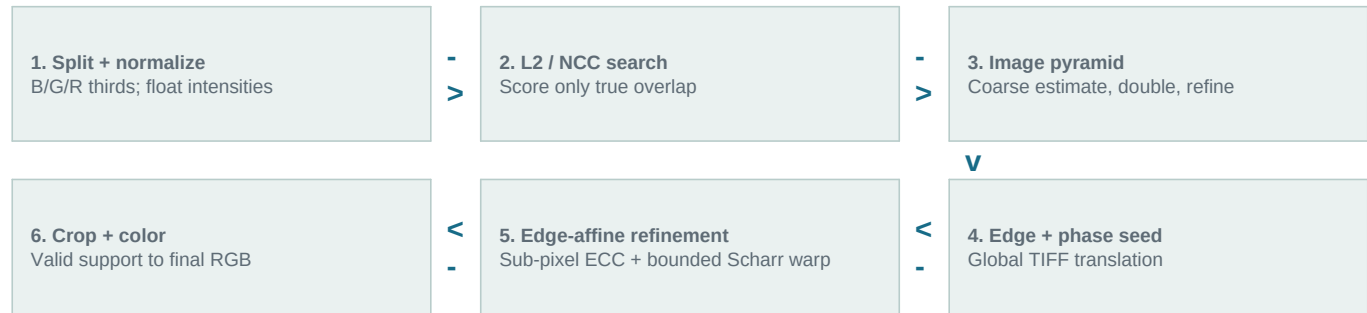


tobolsk - exhaustive single-scale result

2 Multi-Scale Alignment

Large TIFF scans require shifts far outside the JPEG window, and a native-resolution exhaustive search is unnecessarily expensive. The final method therefore evolves from a flat metric search into a coarse-to-fine, edge-aware pipeline.

2.1 Algorithm evolution and final flow



stage	method	reason for the next stage
A	Single-scale L2 / NCC	Simple baseline; reliable for small JPEG displacements
B	Custom 2x pyramid	Propagates $d_l = 2 d_{l-1} + \delta_l$; makes large shifts tractable
C	Edge magnitude + intensity NCC	Shared geometry survives filter-dependent brightness
D	FFT phase seed + bounded local search	Finds the global TIFF shift, then removes thumbnail quantization

2.2 Metric comparison

signal	optimization	advantage	final use
Normalized L2	minimize standardized RMSE	Simple numerical error	Quality audit; nonlinear brightness can hurt
Intensity NCC	maximize normalized dot product	Invariant to mean and scale	Primary local similarity
Edge NCC	maximize edge-pattern agreement	Emphasizes shared structure	30% of the integer-search score
Bounded edge-affine ECC	maximize clipped Scharr-edge agreement	Removes residual scale/rotation/shear fringes	Accepted only after strict quality + deformation gates
Phase correlation	maximize FFT cross-power peak	Fast global translation	Initializer only; local NCC validates it

Final registration: integer search uses $0.70 * \text{NCC}(\text{intensity}) + 0.30 * \text{NCC}(\text{edge magnitude})$. Correct-sign intensity ECC then removes sub-pixel quantization; clipped Scharr-edge ECC may apply a tiny affine residual. The affine candidate is accepted only if center quality rises by at least 0.01, singular values remain in $[0.985, 1.015]$, shear stays below 0.012, and corner deformation is bounded. No image-specific displacement is entered manually.

2.3 Multi-Scale Results - All 14 Course Samples

The table documents the complete supplied dataset, not only the cover images. Positive dy moves a plate downward and positive dx moves it right. NCC is higher-is-better; normalized L2/RMSE is lower-is-better.

image	G -> B	R -> B	NCC G / R	L2 G / R	mean NCC / L2	crop t,b,l,r
cathedral	(5, 2)	(12, 3)	0.96073 / 0.90644	0.28024 / 0.43257	0.93359 / 0.35641	(17, 331, 23, 368)
church	(24, 3)	(57, -4)	0.91496 / 0.75723	0.4124 / 0.6968	0.8361 / 0.5546	(91, 3158, 136, 3454)
emir	(49, 23)	(106, 40)	0.71845 / 0.26693	0.7504 / 1.21084	0.49269 / 0.98062	(169, 3071, 303, 3492)
harvesters	(59, 17)	(123, 15)	0.82266 / 0.82769	0.59555 / 0.58704	0.82518 / 0.59129	(221, 3158, 219, 3412)
icon	(41, 18)	(89, 23)	0.87153 / 0.6841	0.50689 / 0.79486	0.77782 / 0.65087	(120, 3168, 280, 3529)
ilemselga	(41, 7)	(132, 10)	0.74608 / 0.58783	0.71263 / 0.90793	0.66696 / 0.81028	(163, 3222, 465, 3512)
melons	(83, 9)	(178, 12)	0.96216 / 0.92843	0.27511 / 0.37834	0.94529 / 0.32673	(293, 3212, 231, 3459)
monastery	(-3, 2)	(3, 2)	0.94115 / 0.805	0.34307 / 0.6245	0.87308 / 0.48378	(18, 311, 29, 369)
religious_painting	(27, 4)	(68, 7)	0.88024 / 0.80454	0.48941 / 0.62524	0.84239 / 0.55732	(405, 2964, 213, 3450)
self_portrait	(77, 29)	(175, 37)	0.85576 / 0.81991	0.53711 / 0.60016	0.83783 / 0.56863	(219, 3163, 410, 3532)
siren	(49, -5)	(96, -23)	0.8673 / 0.90634	0.51516 / 0.4328	0.88682 / 0.47398	(223, 3160, 286, 3542)
three_generations	(50, 15)	(109, 13)	0.96311 / 0.91594	0.27162 / 0.41002	0.93953 / 0.34082	(261, 3163, 231, 3514)
tobolsk	(3, 2)	(6, 3)	0.91931 / 0.85779	0.40172 / 0.53332	0.88855 / 0.46752	(13, 326, 30, 372)
wharf	(15, -7)	(81, -15)	0.91143 / 0.80401	0.42088 / 0.62608	0.85772 / 0.52348	(176, 3203, 194, 3427)

The fixed method is deliberately global: low scores on a scene with large sky, water, or historical exposure differences are reported rather than hidden. The visual outputs remain the primary reconstruction result; the metrics make the search auditable.

3 Automatic Cropping (Bells & Whistles)

The valid B/G/R intersection is computed from the measured shifts. Within that support, border rows and columns are rejected when they are clipped black/white, low-information, or unusually inconsistent across channels. A consecutive-good-run test prevents one deceptively clean strip from ending the crop too early.

Each crop box is recorded in results.json and section 2.3; no per-image coordinates are hard-coded.



Before: aligned scan retaining plate margins



After: detected common photographic region

4 Automatic Contrasting (Bells & Whistles)

After border removal, each RGB channel is independently mapped from its 1st and 99th percentiles into $[0,1]$. Robust percentiles recover useful shadow and highlight detail without allowing a dust speck, scan border, or saturated pixel to control the full range.

The same percentile rule is used for all course and LOC images. The right result also includes the subsequent fixed white-balance and gamma stages.



Before: aligned and cropped



After: percentile contrast and final color cleanup

5 Automatic White Balance (Bells & Whistles)

Following contrast normalization, restrained gray-world balancing moves the three channel means toward their common mean. Gains are clipped to $[0.85, 1.18]$, preventing noise amplification or implausible neutralization on sparse and nearly monochromatic scenes. A mild gamma of 0.96 follows.

No sample receives hand-selected gains; the same bounded correction is applied to every output.



Before: aligned/cropped input



After: bounded gray-world balance

6 Better Color Mapping (Bells & Whistles)

The source plates are ordered B/G/R, while image encoders expect R/G/B. Alignment and overlap bookkeeping remain in source order, then the aligned stack is explicitly reordered before JPEG encoding. Percentile normalization and bounded gray-world gains supply one deterministic display mapping for every scene.

This avoids a plausible-looking but incorrect red/blue swap; no image uses a hand-picked permutation or color curve.



Aligned/cropped display RGB



Final mapped RGB rendering

7 Better Features (Bells & Whistles)

Equal physical points can have very different raw intensity under the historical filters. The global stage therefore combines intensity NCC, edge NCC, and a Hann-windowed phase-correlation seed. A corrected-sign sub-pixel ECC pass follows. Finally, clipped Scharr edges drive a bounded affine ECC residual that can correct slight glass-plate scale, rotation, and shear differences.

The quality gate accepts affine refinement only when the center score improves by at least 0.01 and the transform stays close to rigid translation. It is accepted for both channels of three_generations and harvesters, reducing thin edge color fringes while preserving genuine subject motion between exposures. Section 2.3 reports metrics computed from the final warped channels.



Aligned/cropped result before color cleanup



Final edge-aware reconstruction

2.4 Multi-Scale Results - Additional Library of Congress Examples

Deliverable requirement: results on 3+ self-selected examples downloaded from the Prokudin-Gorskii collection. These are three original LOC 'digital file from glass neg.' vertical TIFF resources. They were downloaded, split into B / G / R thirds, and processed by exactly the same code and parameters as the 14 course samples.



V lasnoe polianie



V lasnoe polianie (variant)



Na Uralie

item	title	NCC G / R	L2 G / R	mean NCC / L2	offsets G / R
LOC 2018679023	V lasnoe polianie	0.77135 / 0.81494	0.67624 / 0.60837	0.79314 / 0.64231	[42, -6] / [102, 7]
LOC 2018679024	V lasnoe polianie (variant)	0.80382 / 0.77942	0.62638 / 0.6642	0.79162 / 0.64529	[38, 6] / [73, 14]
LOC 2018679025	Na Uralie	0.70154 / 0.83208	0.77261 / 0.57952	0.76681 / 0.67606	[41, 34] / [95, 58]

LOC item 2018679023: <https://www.loc.gov/item/2018679023/>

Raw plate: <https://tile.loc.gov/storage-services/master/pnp/prok/00100/00159u.tif>

LOC item 2018679024: <https://www.loc.gov/item/2018679024/>

Raw plate: <https://tile.loc.gov/storage-services/master/pnp/prok/00100/00160u.tif>

LOC item 2018679025: <https://www.loc.gov/item/2018679025/>

Raw plate: <https://tile.loc.gov/storage-services/master/pnp/prok/00100/00161u.tif>

Appendix A - Visual Gallery (reference only)

The HTML page provides clickable full-size images. This appendix confirms that all 14 course outputs were generated by the same run and keeps the report self-contained for PDF review.



melons



three_generations



cathedral



tobolsk



siren



monastery



wharf



religious_painting



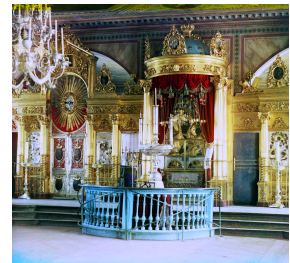
self_portrait



church



harvesters



icon



ilemselga



emir

Source and reproducibility

Course inputs are the supplied CS180_fa2026_proj1_data files. The three extra plates are linked above to LOC item pages and exact TIFF resources. The implementation is code/main.py; the browser report is web/index.html; the Gradescope artifact is this PDF.